SYSDESIGN REFERENCE

Fault Tolerance Tradeoff Matrix

A decision-oriented comparison of alternatives and failure modes.

Keep critical behavior within acceptable limits when dependencies, nodes, or regions fail.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / FAULT-TOLERANCE

SYSDESIGN

Fault Tolerance Tradeoff Matrix

CORE IDEA

Keep critical behavior within acceptable limits when dependencies, nodes, or regions fail.

Reference architecture decision chain

01	02	03	04
Timeouts	Retries	Circuit Breakers	Bulkheads

Decisions to defend

+ Timeouts
+ retries
+ circuit breakers
+ bulkheads
+ redundancy
+ and graceful degradation.

Failure modes to test

+ Retry storms
+ correlated failure
+ false redundancy
+ and silent data loss.

Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

Fault Tolerance Tradeoff Matrix

Probe	What a strong answer makes explicit
How does the design behave during a regional dependency failure?	Requirement, decision, cost, failure behavior, and observable metric.
Which metric reveals the first bottleneck?	Requirement, decision, cost, failure behavior, and observable metric.
What consistency does the user actually observe?	Requirement, decision, cost, failure behavior, and observable metric.
How would the design change at ten times the traffic?	Requirement, decision, cost, failure behavior, and observable metric.

Data questions

+ Partition key
+ Versioning
+ Uniqueness
+ Retention
+ Reconciliation

Operations questions

+ SLO
+ Backlog
+ Capacity
+ Alert
+ Failover test

DESIGN DISCIPLINE

For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?

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